

Incident Response Report

Executive Summary

A simulated cyber security incident was conducted within a controlled virtual lab environment using Kali Linux, Metasploitable2, Wazuh SIEM, Metasploit Framework, Docker, and CrowdSec. The objective was to demonstrate the complete Security Operations Center (SOC) workflow including attack simulation, monitoring, detection, containment, and forensic evidence preservation. A Samba vulnerability on the target system was successfully exploited using the Metasploit Framework, resulting in unauthorized remote shell access.

Timeline

- Environment configured using VirtualBox with Host-Only networking.
- Vulnerable Samba service identified using Nmap and Metasploit.
- Exploit executed against Metasploitable2 target system.
- Reverse shell access successfully established.
- Wazuh SIEM deployed using Docker containers.
- Security alerts monitored through the Wazuh dashboard.
- CrowdSec used to block the attacker IP address.
- SHA256 hashing performed for forensic evidence preservation.

Recommendations

- Disable unnecessary network services such as vulnerable Samba versions.
- Apply security patches and updates regularly.
- Deploy centralized SIEM monitoring for real-time threat detection.
- Implement network segmentation and firewall rules.
- Use threat intelligence frameworks such as MITRE ATT&CK.
- Maintain forensic evidence handling procedures during incidents.